

Amazon River Navigation

Technical Planning Document

CBC AI Builders Hackathon | Illinois Institute of Technology

Build: 2:10–4:30 PM | Demo: 4:30–5:30 PM

Problem

Seasonal flooding in the Amazon basin creates and destroys navigable routes every year. During high water, shortcuts emerge through flooded forest (várzea) that experienced locals navigate from memory but that exist in no digital map. During low water, those same channels vanish. Google Maps, ANA, and INPE provide static or monitoring data — none generate dynamic routing based on seasonal water presence. Municipal governments and logistics operators have no tool to query these conditions programmatically.

Data Foundation

JRC Global Surface Water (Google / European Commission, public domain)

Layer	Contents	URL pattern
Occurrence	% of months 1984–2021 each pixel had water	occurrence_{TILE}v1_4_2021.tif
Monthly History	Binary water/no-water per pixel per month since 1984	monthlyHistory_{TILE}_{YEAR}_{MONTH}v1_4_2021.tif

Resolution: 30 m/pixel (Landsat). Pixels are classified into four categories based on occurrence value:

Category	Threshold	Meaning
Permanent water	> 90%	Navigable year-round
Seasonal water	15–90%	Month-dependent navigability
Rare flood	< 15%	Caution; unreliable
Land	0%	Not navigable

Validated tile: `occurrence_70W_0Nv1_4_2021.tif` covers my island at `(-3.338949, -60.189)`, which shows clear seasonal classification in the occurrence layer.

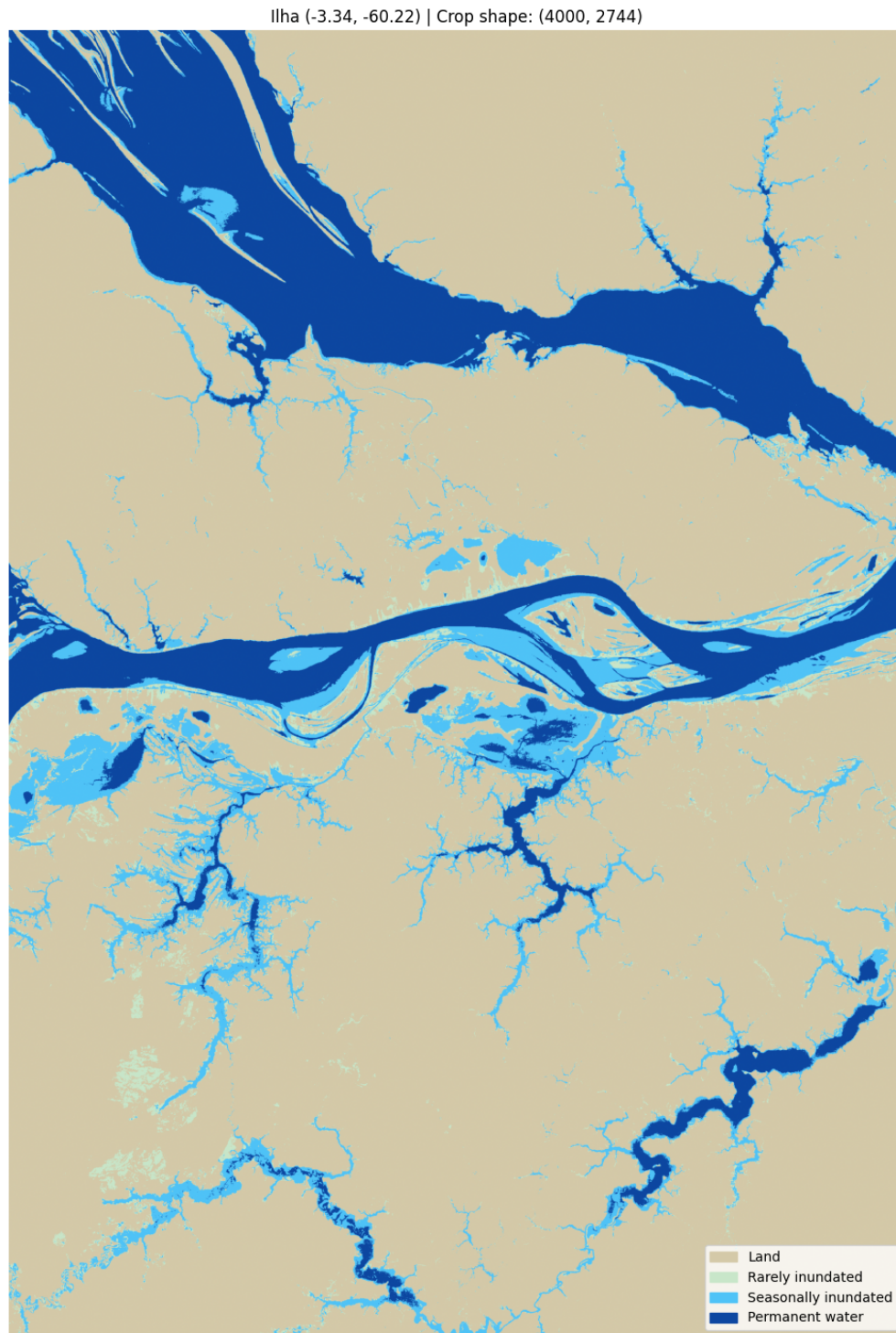


Figure 1: Example of a region in the Amazon made with the database

Architecture

```
[JRC GeoTIFF tiles -- occurrence + monthly history]
|
[Flask backend -- rasterio pixel processing]
  extract seasonal pixels by month/bbox
  |
[Route logic -- connected seasonal pixel graph]
  identify navigable corridors above threshold
  |
[Claude API -- claude-sonnet-4-20250514]
  translate pixel data into plain-language advisory
  |
[Leaflet.js frontend]
  interactive map + month slider + advisory panel
```

Stack

Layer	Technology	Rationale
Data	JRC Global Surface Water	Public, free, satellite-validated, monthly granularity
Backend	Python + Flask + rasterio	GeoTIFF processing, bbox queries, pixel classification
AI	Claude API (Sonnet 4)	Converts scientific pixel data into actionable advisory
Frontend	Leaflet.js	Lightweight tiled map, well-suited for overlay rendering
Deploy	Local / ngrok for demo	Hackathon scope

Core Demo Features

Month slider — drag from January to December, watch routes appear and disappear as seasonal pixels activate and deactivate across the map.

Navigability overlay — three rendered layers: permanent water (always navigable), seasonal corridors (month-dependent), and land. Color-coded for instant readability.

Claude advisory panel — on each slider change, the backend sends pixel statistics for the selected month to Claude, which returns a 2–3 sentence plain-language summary: current conditions, which corridors are open, and any caution flags.

AI Integration

Claude is not decorative. The raw data output is a pixel count and a classification array. No municipal logistics officer can act on that. Claude's role is **translation between scientific data and operational decision-making**.

Input sent to Claude (per query)

```
Month: March
Seasonal pixels active in corridor A: 847 (up from 312 in January)
Seasonal pixels active in corridor B: 203 (down from 891 in November)
Permanent water pixels: 14,203 (unchanged)
```

Claude output (shown in advisory panel)

```
In March, the northern floodplain corridor near [area] is fully
navigable
-- water levels are at seasonal peak. The southern channel has
significantly
less water presence than November and should be treated with caution for
vessels with draft above 0.5m. Permanent river routes remain unaffected.
```

This pipeline directly addresses the hackathon criterion *Best Use of AI Workflow*: Claude sits between satellite data and operational decision-making, not alongside it.

Known Limitations

Historical data only. Occurrence data covers 1984–2021, not real time. Production integration would use ANA (Agência Nacional de Águas) real-time gauge data.

Surface water, not bathymetry. Routes are approximated from water presence, not depth measurements. Draft calculations are estimates only.

Climatological averages. Monthly history averages across all years; a drought year like 2010 is averaged with normal years. Interannual variability is not yet captured. We can compute, for each pixel and each calendar month, the 25th percentile, median, and 75th percentile across the 37 annual observations available in the Monthly History layer. This converts a single average into a confidence interval, allowing the Claude advisory layer to report not just expected conditions but the realistic range — "under normal conditions this corridor is navigable in March; in drought years consistent with negative ENSO phases, water presence drops below the navigability threshold."

No vessel profile input. All routes currently assume small-craft navigation. Draft and beam input is a planned feature.

Hackathon Alignment

Criterion	Weight	How we address it
Usefulness	30%	Real operator need, defined user (municipal logistics), no existing solution
Build Quality	25%	Live map with satellite data, functional slider, real API calls
Use of AI	25%	Claude translates satellite pixels into operational language
Clarity & Responsibility	20%	Limitations disclosed upfront; personal story establishes genuine context

Theme fit — Economic Empowerment: reliable seasonal navigation directly reduces logistics cost and emergency response time in the most underserved geography in South America.

Team

Member	Role
Gabriel	Data domain, Claude prompt engineering, full pitch
Danish	Flask backend, Claude API integration, GeoTIFF processing
Ayaan	Leaflet.js frontend, month slider, AI workflow documentation